Commonly Asked Git & Linux MCQs for Placements

Placement-oriented practice set • Git + Linux fundamentals, commands, workflows and troubleshooting

How to use this document: Attempt each question first without looking at the answer key. The questions emphasize concepts frequently tested in technical placement assessments, especially for software, embedded, DevOps and systems-oriented roles.

Part A — Git MCQs

51 placement-focused questions covering Git basics, staging, branches, merge/rebase, reset/revert, remotes, stash, tags and collaboration.

1. Which command initializes a new Git repository in the current directory?

A. git start
B. git init
C. git create
D. git new

2. Which command displays the current state of the working tree and staging area?

A. git status
B. git state
C. git check
D. git diff-status

3. Which command stages a file for the next commit?

A. git add
B. git stage-only
C. git commit
D. git track

4. Which command creates a commit from the staged changes?

A. git save
B. git commit
C. git push
D. git snapshot

5. Which command uploads local commits to a remote repository?

A. git pull
B. git fetch
C. git push
D. git upload

6. Which command downloads remote changes and integrates them into the current branch?

A. git pull
B. git fetch
C. git clone
D. git sync

7. Which command downloads remote references without integrating them into the current branch?

A. git pull
B. git fetch
C. git clone
D. git update

8. Which command creates a local copy of an existing remote repository?

A. git fork
B. git clone

C. git copy
D. git mirror

9. Which command lists local branches?

A. git branch
B. git list
C. git branches
D. git show-branch-only

10. Which command creates and switches to a new branch in traditional Git syntax?

A. git branch -s feature
B. git checkout -b feature
C. git switch feature -n
D. git new feature

11. Which modern command switches to an existing branch?

A. git switch
B. git move
C. git branch --use
D. git checkout-only

12. Which command combines one branch's history into the current branch using a merge commit when necessary?

A. git merge
B. git join
C. git combine
D. git integrate

13. What is the purpose of git rebase?

A. Delete all branches
B. Move/replay commits onto a new base
C. Upload tags
D. Compress files

14. What is a fast-forward merge?

A. A merge with conflicts
B. A merge where the branch pointer can simply advance
C. A merge using rebase
D. A merge that deletes history

15. Which command shows commit history?

A. git history
B. git log
C. git commits
D. git timeline

16. Which command shows the differences between working-tree changes and the last commit/staged state as applicable?

A. git diff
B. git compare
C. git changes
D. git delta

17. Which command shows changes that are staged for commit?

A. git diff --cached
B. git staged
C. git diff --stage
D. git show --staged-only

18. Which command removes a file from the staging area while keeping the file's working-tree changes?

A. git reset HEAD <file>

B. git delete <file>
C. git clean <file>
D. git remove-stage <file>

19. Which command removes untracked files from the working tree?

A. git clean
B. git purge
C. git remove-untracked
D. git reset --hard

20. Which command creates a lightweight or annotated reference to a specific point in history?

A. git tag
B. git mark
C. git label
D. git point

21. What is HEAD in Git?

A. The remote repository
B. The current checked-out commit/reference
C. The staging area
D. The first commit only

22. What is the staging area also called?

A. Index
B. Workspace
C. HEAD
D. Object cache

23. Which area contains files as they exist in the currently checked-out version?

A. Working tree
B. Remote
C. Object database only
D. Stash

24. Which command temporarily stores uncommitted changes?

A. git stash
B. git hold
C. git suspend
D. git temp

25. Which command lists stashed entries?

A. git stash list
B. git stash show-all
C. git list stash
D. git stash --all

26. Which command applies the latest stash and removes it from the stash list?

A. git stash pop
B. git stash apply
C. git stash use
D. git stash restore

27. What does git revert do?

A. Deletes a commit from all history
B. Creates a new commit that reverses an earlier commit
C. Removes the repository
D. Resets only the remote branch

28. What does git reset --hard primarily do?

A. Only changes the remote repository
B. Moves HEAD and resets index and working tree to the target

C. Creates a new revert commit
D. Only unstages files

29. Which reset mode changes HEAD and the index but generally keeps working-tree changes?

A. --soft
B. --mixed
C. --hard
D. --keep-index

30. Which reset mode moves HEAD but leaves both index and working tree unchanged?

A. --soft
B. --mixed
C. --hard
D. --preserve

31. What is a merge conflict?

A. A Git installation error
B. A situation where Git cannot automatically reconcile changes
C. A network timeout
D. A corrupted .git folder

32. Which command helps identify the commit that introduced a particular line?

A. git blame
B. git inspect
C. git author
D. git line-log

33. Which command shows information about a particular commit?

A. git show
B. git inspect-commit
C. git details
D. git commit-info

34. What is a remote-tracking branch such as origin/main?

A. A local branch stored only on another PC
B. A local reference tracking a branch on a remote
C. A temporary stash
D. A tag

35. What does origin usually represent?

A. The default remote name
B. The root commit
C. The staging area
D. The first branch

36. Which command lists configured remotes and their URLs?

A. git remote -v
B. git remote-list
C. git url
D. git config --remote

37. What is the main purpose of .gitignore?

A. Ignore Git itself
B. Specify untracked files/patterns Git should normally ignore
C. Delete remote files
D. Hide commit history

38. If a file is already tracked, adding it to .gitignore will automatically stop tracking it. True or False?

A. True
B. False
C. Only on GitHub

D. Only after push

39. Which command creates a new local branch at the current commit without switching to it?

A. git branch feature
B. git switch feature
C. git checkout feature
D. git branch --switch feature

40. Which command deletes a local branch after it has been merged?

A. git branch -d feature
B. git delete feature
C. git remove-branch feature
D. git branch --rm feature

41. Which command displays the configured Git user name?

A. git user
B. git config user.name
C. git whoami
D. git identity

42. Which command creates a new commit even if there are no changes, when allowed?

A. git commit --allow-empty
B. git commit --force-empty
C. git empty-commit
D. git commit -0

43. What is the main purpose of a Git remote?

A. To provide a reference/location for another repository
B. To compile code
C. To stage files
D. To replace branches

44. Which statement about Git is correct?

A. Git requires a central server for every commit
B. Git is a distributed version-control system
C. Git stores only the latest file version
D. Git cannot work offline

45. Which command can create an annotated tag?

A. git tag -a v1.0 -m "release"
B. git tag --commit v1.0
C. git release v1.0
D. git mark -a v1.0

46. Which operation rewrites existing commit history?

A. git rebase
B. git status
C. git fetch
D. git diff

47. Why should force-pushing rewritten history to a shared branch be avoided?

A. It increases file size
B. It can discard or invalidate collaborators' published history
C. It prevents cloning forever
D. It disables Git tags

48. Which command is commonly used to find a commit by searching commit messages?

A. git log --grep
B. git find-message
C. git search-log
D. git commit --find

49. Which command shows the branches containing a particular commit?

A. git branch --contains <commit>
B. git branch --commit <commit>
C. git contains <commit>
D. git show-branches <commit>

50. What is a pull request primarily used for?

A. Requesting review and integration of proposed changes
B. Downloading Git itself
C. Creating a local repository
D. Replacing commits

51. Which is the best description of GitHub/GitLab/Bitbucket relative to Git?

A. They are implementations of the Git command only
B. They are hosting/collaboration platforms that can use Git repositories
C. They replace the Git object model
D. They are Linux shells

Part B — Linux MCQs

53 placement-focused questions covering shell commands, filesystems, permissions, processes, redirection, pipes, environment variables and basic system administration.

1. Which command prints the current working directory?

A. pwd
B. cwd
C. where
D. dir

2. Which command lists files in a directory?

A. ls
B. list
C. dirlist
D. show

3. Which option commonly shows hidden files with ls?

A. -h
B. -a
C. -r
D. -x

4. Which command changes the current directory?

A. cd
B. chdir-only
C. move
D. goto

5. Which command creates a new directory?

A. mkdir
B. mkfile
C. newdir
D. diradd

6. Which command creates an empty file if it does not exist and updates its timestamp if it does?

A. touch
B. mkfile
C. create
D. file

7. Which command copies files?

A. cp
B. copyfile
C. mv
D. clone

8. Which command moves or renames a file?

A. mv
B. rn
C. move-only
D. shift

9. Which command removes a file?

A. rm
B. del
C. erase-file
D. remove-file

10. Which option is commonly required to remove a directory recursively with rm?

A. -r
B. -p
C. -d
D. -s

11. Which command displays the contents of a text file directly?

A. cat
B. readfile
C. show
D. typefile

12. Which command is commonly used to view a long file one screen/page at a time?

A. less
B. cat -page
C. screenfile
D. morefile-only

13. Which command searches for matching text patterns in files?

A. grep
B. findtext
C. scan
D. search

14. Which command finds files/directories based on criteria such as name?

A. find
B. locate-file-only
C. searchfs
D. whereis-all

15. Which command displays running processes in a snapshot?

A. ps
B. proc
C. jobs-only
D. runlist

16. Which command provides an interactive real-time process view on many Linux systems?

A. top
B. ps -live
C. procwatch
D. liveps

17. Which command sends a signal to a process?

A. kill

B. stop
C. signal
D. terminate

18. What is the purpose of PID?

A. Identify a process
B. Identify a file
C. Identify a user
D. Identify a shell command

19. Which command changes file permissions?

A. chmod
B. chperm
C. perm
D. setmode

20. Which command changes file ownership?

A. chown
B. owner
C. chmod-owner
D. own

21. Which command changes group ownership?

A. chgrp
B. groupmod-file
C. chmod-g
D. setgroup

22. In rwx permissions, what does r represent?

A. Read
B. Run
C. Root
D. Rename

23. In rwx permissions, what does w represent?

A. Write
B. Wait
C. Watch
D. Web

24. In rwx permissions, what does x represent?

A. Execute
B. Extract
C. Extend
D. Export

25. What does chmod 755 file commonly mean?

A. Owner rwx; group r-x; others r-x
B. Owner rw-; group r-- only
C. Everyone rwx
D. Owner r--; others rwx

26. What does chmod 644 file commonly mean?

A. Owner rw-; group r--; others r--
B. Owner rwx; everyone r-x
C. Owner r--; group rw-; others rwx
D. Everyone rw-

27. Which symbol represents the root directory?

A. /
B. ~

C. #
D. .

28. Which symbol usually represents the current user's home directory in the shell?

A. ~
B. /
C. .
D. ..

29. Which path represents the parent directory?

A. ..
B. .
C. ~
D. /..

30. Which path represents the current directory?

A. .
B. ..
C. ./..
D. ~

31. What is an absolute path?

A. A path beginning from the filesystem root or another defined root
B. A path using only filenames
C. A path containing no slash
D. A path relative to the current directory

32. What is a relative path?

A. A path interpreted relative to the current working directory
B. A path beginning with /
C. A path that must contain ~
D. A path used only by root

33. Which command displays disk-space usage of mounted filesystems?

A. df
B. du
C. disk
D. space

34. Which command estimates disk usage of files/directories?

A. du
B. df
C. size
D. diskuse-only

35. Which command displays memory usage?

A. free
B. mem
C. ram
D. memory

36. Which command displays the last lines of a file?

A. tail
B. end
C. bottom
D. lastlines

37. Which command displays the first lines of a file?

A. head
B. start
C. topfile

D. first

38. Which command is commonly used to follow a growing log file in real time?

A. tail -f
B. head -f
C. cat -watch
D. grep -live

39. Which operator redirects standard output to a file and overwrites it?

A. >
B. >>
C. <
D. |

40. Which operator appends standard output to a file?

A. >>
B. >
C. <<
D. |

41. Which operator connects stdout of one command to stdin of another?

A. |
B. >
C. >>
D. &

42. Which file descriptor is standard input?

A. 0
B. 1
C. 2
D. 3

43. Which file descriptor is standard output?

A. 1
B. 0
C. 2
D. 3

44. Which file descriptor is standard error?

A. 2
B. 0
C. 1
D. 3

45. Which command prints environment variables?

A. env
B. vars
C. printenv-only-all
D. environment

46. Which command displays a command's executable location when found in PATH?

A. which
B. where
C. locatecmd
D. pathof

47. Which variable stores the executable search path in a typical shell?

A. PATH
B. HOME
C. PWD
D. SHELLPATH

48. Which command shows the exit status of the most recently executed foreground command in many shells?

A. echo $?
B. status
C. echo $STATUS
D. ret

49. What does an exit status of 0 conventionally mean?

A. Success
B. Failure
C. Permission denied
D. Process killed

50. Which command creates a symbolic link?

A. ln -s
B. link -soft
C. symlink
D. ln --symbolic-only

51. Which command shows information about a command, including whether it is a shell builtin/alias/function/executable in many shells?

A. type
B. info-command
C. what
D. describe

52. Which command changes the current user's password?

A. passwd
B. password
C. chpasswd-user-only
D. setpass

53. Which command runs a command with another user's privileges when configured appropriately?

A. sudo
B. suudo
C. admin
D. rootcmd

Answer Key & Quick Explanations — Git

1. B — git init creates a new .git directory and initializes the repository.

2. A — git status shows modified, staged, untracked and other working-tree information.

3. A — git add places changes into the staging area (index).

4. B — git commit records the staged snapshot in the local repository.

5. C — git push transfers local commits/refs to a remote repository.

6. A — git pull normally performs a fetch followed by integration, usually merge or rebase depending on configuration/options.

7. B — git fetch updates remote-tracking references without changing the current branch.

8. B — git clone creates a new local repository based on an existing repository.

9. A — git branch lists local branches by default.

10. B — git checkout -b creates the branch and checks it out. Modern Git also supports git switch -c feature.

11. A — git switch is designed specifically for switching branches.

12. A — git merge integrates another branch into the current branch.

13. B — Rebase rewrites commit ancestry by replaying commits on top of another base commit.

14. B — If the current branch is an ancestor of the branch being merged, Git can move the pointer forward without creating a merge commit.

15. B — git log displays commit history.

16. A — git diff is used to inspect differences; without options it generally compares working tree with the index.

17. A — git diff --cached (or --staged) compares the index with HEAD.

18. A — git reset HEAD <file> (or git restore --staged <file>) unstages the file without normally discarding its working-tree content.

19. A — git clean removes untracked files/directories according to its options. It can be destructive.

20. A — git tag creates tags, commonly used to mark releases.

21. B — HEAD identifies the currently checked-out commit, normally through the current branch reference.

22. A — Git's staging area is commonly called the index.

23. A — The working tree is the set of checked-out files you edit.

24. A — git stash saves local modifications and returns the working tree to a cleaner state.

25. A — git stash list displays the stash stack.

26. A — git stash pop applies the stash and removes it if successful. git stash apply keeps it.

27. B — git revert is history-preserving: it creates a new commit that undoes the effect of a previous commit.

28. B — It resets HEAD, the index and working tree to the specified commit, potentially discarding uncommitted work.

29. B — git reset --mixed (the default) moves HEAD and resets the index while keeping working-tree changes.

30. A — git reset --soft changes HEAD only, leaving staged and working-tree content intact.

31. B — Conflicts occur when Git cannot automatically determine how overlapping changes should be combined.

32. A — git blame annotates lines with the commit and author information associated with their latest change.

33. A — git show displays commit metadata and associated changes.

34. B — origin/main is a remote-tracking reference representing the last known state of main on remote origin.

35. A — origin is the conventional default name assigned to the remote from which a repository was cloned.

36. A — git remote -v shows remote names and fetch/push URLs.

37. B — A .gitignore file defines patterns for files Git should not normally consider as untracked.

38. B — Gitignore does not untrack an already tracked file; you must remove it from the index, e.g. git rm --cached.

39. A — git branch feature creates the branch pointer but leaves the current branch checked out.

40. A — git branch -d safely deletes a local branch when Git considers it merged.

41. B — git config user.name reads the configured user.name value.

42. A — git commit --allow-empty creates an empty commit.

43. A — A remote stores the connection information for another Git repository.

44. B — Git is distributed: repositories contain substantial history and most operations can be performed locally.

45. A — git tag -a creates an annotated tag; -m supplies its message.

46. A — Rebase rewrites commit ancestry. Other operations listed do not normally rewrite existing commits.

47. B — Rewriting shared history can make collaborators' local histories diverge and may cause commits to become difficult to recover.

48. A — git log --grep='pattern' filters commits by commit-message text.

49. A — git branch --contains lists branches whose history contains the specified commit.

50. A — A pull request is a collaboration mechanism on hosting platforms for reviewing and merging changes.

51. B — These platforms provide hosting, collaboration, code review, CI/CD and other services around Git repositories.

Answer Key & Quick Explanations — Linux

1. A — pwd means print working directory.

2. A — ls lists directory contents.

3. B — ls -a includes entries whose names begin with a dot.

4. A — cd changes the shell's current working directory.

5. A — mkdir creates directories.

6. A — touch creates an empty file or updates timestamps.

7. A — cp copies files/directories according to its options.

8. A — mv is used for moving and renaming.

9. A — rm removes directory entries/files.

10. A — rm -r recursively removes directories and their contents.

11. A — cat writes file contents to standard output.

12. A — less provides interactive paged viewing and is commonly used for long files.

13. A — grep searches input for lines matching a pattern.

14. A — find recursively searches directory trees using criteria such as -name, -type and -size.

15. A — ps reports current processes.

16. A — top displays processes and resource usage interactively.

17. A — kill sends a signal to a process by PID; despite its name, it can send signals other than SIGKILL.

18. A — PID means process ID and uniquely identifies a process at a given time.

19. A — chmod changes permission bits.

20. A — chown changes the user and/or group ownership of files.

21. A — chgrp changes the group owner of a file.

22. A — r is read permission.

23. A — w is write permission.

24. A — x is execute permission for files and traversal permission for directories.

25. A — 7=rwx, 5=r-x, 5=r-x.

26. A — 6=rw-, 4=r--, 4=r--.

27. A — The filesystem root is /.

28. A — The tilde expands to the current user's home directory in typical shells.

29. A — .. refers to the parent directory.

30. A — . refers to the current directory.

31. A — An absolute path identifies a location independently of the current working directory, e.g. /home/user/file.

32. A — Relative paths are interpreted from the current working directory.

33. A — df reports filesystem disk-space usage.

34. A — du reports estimated space used by files/directories.

35. A — free reports free and used memory and related statistics.

36. A — tail prints the end of files.

37. A — head prints the beginning of files.

38. A — tail -f follows appended data as a file grows.

39. A — > redirects stdout and truncates/overwrites the destination file unless shell-specific conditions intervene.

40. A — >> appends stdout to the destination file.

41. A — The pipe operator sends standard output to the next command's standard input.

42. A — File descriptor 0 is stdin; 1 is stdout and 2 is stderr.

43. A — File descriptor 1 is stdout.

44. A — File descriptor 2 is stderr.

45. A — env prints the environment; printenv is another common command for this purpose.

46. A — which commonly reports the path of an executable resolved through PATH.

47. A — PATH contains directories searched for executable commands.

48. A — $? expands to the previous command's exit status in common shells.

49. A — Unix/Linux commands conventionally return 0 for success and a nonzero value for failure or other conditions.

50. A — ln -s creates a symbolic (soft) link.

51. A — type is a shell builtin in common shells and identifies how a command name is resolved.

52. A — passwd is used to change passwords, subject to system policy.

53. A — sudo executes commands with privileges authorized by sudo policy.

High-Yield Placement Revision Sheet

Topic	Remember
Git workflow	Working tree → staging area (index) → local repository → remote repository
Core Git commands	git init, git status, git add, git commit, git log, git diff, git branch, git switch, git merge, git rebase, git fetch, git pull, git push
Merge vs rebase	Merge combines histories and may create a merge commit. Rebase replays commits on a new base and rewrites history.
Reset vs revert	reset moves references/history position and can alter the working state; revert creates a new commit that undoes an earlier commit.
Fetch vs pull	fetch downloads remote references without integrating; pull normally fetches and then integrates.
Stash	Temporarily stores uncommitted work so you can switch context; pop applies and removes the stash.
HEAD	Reference to the currently checked-out commit; normally points through the current branch.
Gitignore	Prevents specified untracked files/patterns from normally appearing as untracked; it does not untrack already tracked files.
Linux filesystem	/ is root, ~ is the current user's home, . is current directory, .. is parent directory.
Permissions	r=4, w=2, x=1. Therefore 755=rwx/r-x/r-x and 644=rw-/r--/r--.
Standard streams	stdin=0, stdout=1, stderr=2.
Redirection	> overwrites, >> appends, | pipes stdout to another command's stdin.
Processes	ps gives a process snapshot, top gives an interactive process view, kill sends a signal.
Disk/memory	df=filesystem free space, du=directory/file disk usage, free=memory statistics.
Search	grep searches text patterns; find searches filesystem objects by criteria.
Exit status	0 conventionally indicates success; nonzero indicates failure or another condition.
Links	ln -s creates a symbolic link.
Privilege	sudo runs a command with authorized elevated privileges; use carefully.

Placement tip: Do not memorize commands in isolation. Placement tests often give a command sequence or a small terminal output and ask what the final state will be. Practice predicting the result of git add/commit/push, merge vs rebase, chmod values, pipes/redirection, and process commands.